A Comparative and Explainable Machine Learning Approach for Predicting Heart Disease

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1. Introduction

Cardiovascular diseases (CVDs) remain the leading cause of mortality worldwide, accounting for approximately 19.8 million deaths annually, or nearly 32% of all global deaths [1]. Heart disease, in particular, poses a significant burden on healthcare systems, especially in low- and middle-income countries where access to early diagnostic tools may be limited. Early identification of individuals at risk plays a crucial role in preventing disease progression and reducing mortality.

Recent advancements in healthcare data collection and electronic medical records have enabled the application of machine learning (ML) techniques to cardiovascular risk prediction. Unlike traditional statistical approaches, ML models are capable of capturing complex, non-linear relationships among multiple clinical variables, thereby improving predictive performance [1]. However, despite their accuracy, many ML models function as black boxes, limiting their adoption in clinical practice due to a lack of interpretability.

To address this challenge, Explainable Artificial Intelligence (XAI) techniques such as SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations) have been introduced to improve transparency and trust in ML-driven clinical decision support systems [1]. These methods allow clinicians and researchers to understand both global feature importance and local prediction reasoning.

This study presents a comparative and explainable machine learning framework for predicting heart disease using structured clinical data. Beyond predictive performance, the study emphasizes model interpretability, responsible deployment, and robustness to distributional limitations. An end-to-end pipeline is developed, encompassing data preprocessing, model training, hyperparameter optimization, explainability analysis, and interactive deployment with explicit handling of out-of-distribution (OOD) inputs.

1.1 Objectives

1.1.1 To develop and compare multiple supervised machine learning models for heart disease prediction using clinically relevant attributes.
1.1.2 To optimize model performance with a particular focus on recall, minimizing false negatives in a medical screening context.
1.1.3 To apply explainable AI techniques (SHAP and LIME) to enhance global and local interpretability of model predictions.
1.1.4 To design and deploy an interactive prediction system with responsible handling of out-of-distribution inputs.
1.1.5 To evaluate the limitations of the dataset and propose directions for future clinical and methodological improvements.

2. Methodology

2.1 Research Design

This study adopts a **predictive, quantitative research design** aimed at developing, evaluating, and explaining supervised machine learning models for heart disease prediction. The research emphasizes both **predictive performance** and **model interpretability**, recognizing the critical importance of transparency in medical decision-support systems. The methodology integrates conventional machine learning workflows with explainable artificial intelligence (XAI) techniques and interactive deployment, forming a complete end-to-end predictive pipeline.

The overall process follows a structured sequence consisting of data acquisition, exploratory data analysis (EDA), preprocessing, model training, evaluation, hyperparameter optimization, explainability analysis, and deployment. **Figure 1** illustrates the complete

methodological workflow adopted in this study.

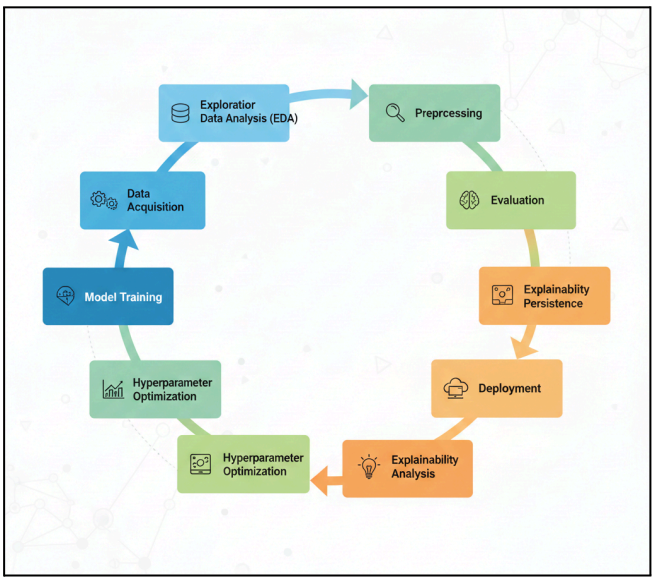


Figure 1. End-to-end machine learning workflow for heart disease prediction, from data acquisition to deployment.

2.2 Dataset Description and Source

The dataset used in this research is the **Heart Failure Prediction Dataset (heart.csv)** obtained from Kaggle [5]. It contains **918 patient records**, each described by **11 clinical attributes** and one binary target variable, **HeartDisease**, indicating the presence (1) or absence (0) of heart disease.

The attributes include a combination of demographic information, physiological measurements, laboratory test results, exercise-induced indicators, and electrocardiographic findings. These variables are commonly used in clinical cardiovascular risk assessment and have been widely adopted in prior machine learning studies.

	Age	Sex	ChestPainType	RestingBP	Cholesterol	FastingBS	RestingECG	MaxHR	ExerciseAngina	Oldpeak	ST_Slope	HeartDisease
0	40	M	ATA	140	289	0	Normal	172	N	0.0	Up	0
1	49	F	NAP	160	180	0	Normal	156	N	1.0	Flat	1
2	37	M	ATA	130	283	0	ST	98	N	0.0	Up	0
3	48	F	ASY	138	214	0	Normal	108	Y	1.5	Flat	1
4	54	M	NAP	150	195	0	Normal	122	N	0.0	Up	0

Table 1. Description of clinical attributes used in the study.

2.3 Exploratory Data Analysis (EDA)

Exploratory Data Analysis was conducted to understand the statistical properties, distributions, and data quality of the features. Initial inspection using df.head(), df.info(), df.describe(), and null-value checks confirmed that the dataset contained no explicit missing values. However, domain-informed analysis revealed the presence of **medically implausible zero values** in certain numerical features.

To visually assess feature distributions, histograms with kernel density estimates were generated for numerical variables, while count plots were used for categorical variables. These visualizations highlighted abnormal spikes at zero for **Resting Blood Pressure** and **Serum Cholesterol**, indicating data quality issues.

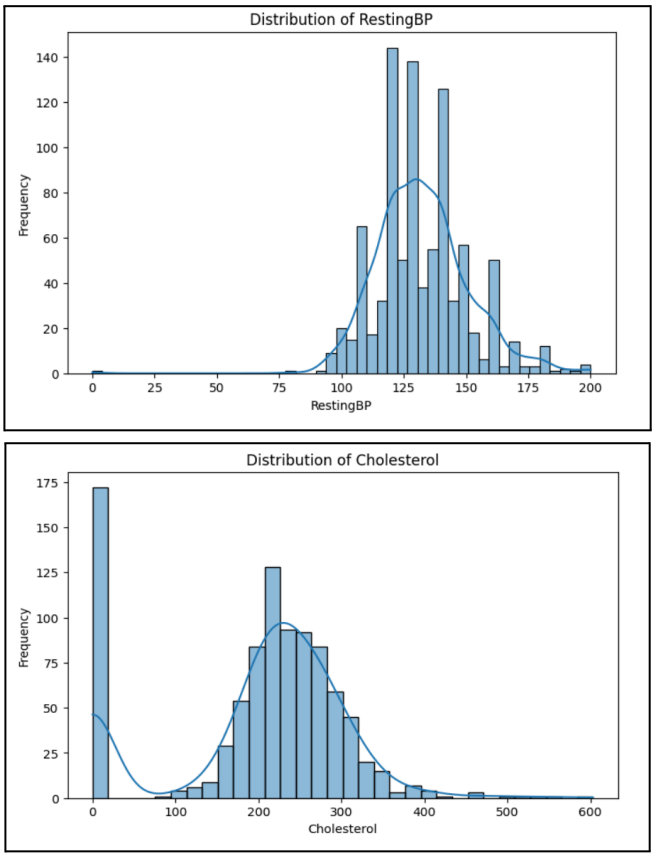


Figure 2. Distribution of numerical features prior to data cleaning.

2.4 Data Cleaning and Handling of Erroneous Values

Medical domain constraints were applied to identify erroneous measurements. Zero values were found in:

- **RestingBP** (1 instance)
- **Cholesterol** (172 instances)

Since zero values for these attributes are physiologically impossible, they were treated as erroneous rather than valid measurements. To correct this, zero values were replaced using the **median of the non-zero values** for each feature (130.0 for RestingBP and 237.0 for Cholesterol). The median was selected to ensure robustness against skewness and outliers.

Following imputation, updated histograms were generated to confirm that the corrected distributions reflected realistic physiological ranges.

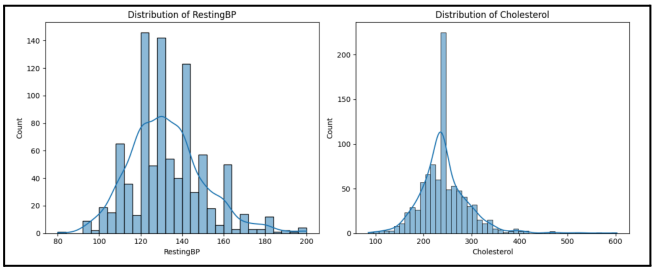


Figure 3. Distribution of RestingBP and Cholesterol after median imputation.

2.5 Data Preprocessing

2.5.1 Feature Encoding

Categorical variables (**Sex, ChestPainType, FastingBS, RestingECG, ExerciseAngina, ST_Slope**) were transformed using **one-hot encoding**. The drop_first=True parameter was applied to prevent multicollinearity by removing redundant dummy variables.

2.5.2 Train–Test Split

The dataset was divided into training and testing subsets using an **80:20 split**, with random_state = 42 to ensure reproducibility. The training set was used exclusively for model fitting and optimization, while the test set was reserved for unbiased evaluation.

```
Data preprocessing complete.
Shape of X_train: (734, 15)
Shape of X_test: (184, 15)
Shape of y_train: (734,)
Shape of y_test: (184,)
```

Figure 4. Data partitioning strategy used for training and testing.

2.5.3 Feature Scaling

Numerical features (**Age, RestingBP, Cholesterol, MaxHR, Oldpeak**) were standardized using **StandardScaler**. The scaler was fitted only on the training data and subsequently applied to both training and test sets to prevent data leakage. Although tree-based models are scale-invariant, scaling was retained for consistency across all evaluated models.

2.6 Model Training and Baseline Evaluation

Five supervised classification models were trained and evaluated:

1. Logistic Regression
2. Support Vector Classifier (SVC)
3. Random Forest Classifier
4. k-Nearest Neighbors (KNN)
5. Gaussian Naïve Bayes

Each model was trained on the preprocessed training data and evaluated on the test set using **accuracy, precision, recall, F1-score, and ROC-AUC**. In line with clinical screening priorities, **recall** was emphasized to reduce false negatives, as undetected heart disease poses significant clinical risk.

	Accuracy	Precision	Recall	F1 Score	ROC-AUC
Logistic Regression	0.8587	0.9091	0.8411	0.8738	0.9247
SVC	0.8370	0.8667	0.8505	0.8585	0.9351
Random Forest	0.8641	0.8942	0.8692	0.8815	0.9220
KNeighbors	0.8533	0.8774	0.8692	0.8732	0.9191
Gaussian Naive Bayes	0.8478	0.9158	0.8131	0.8614	0.9226

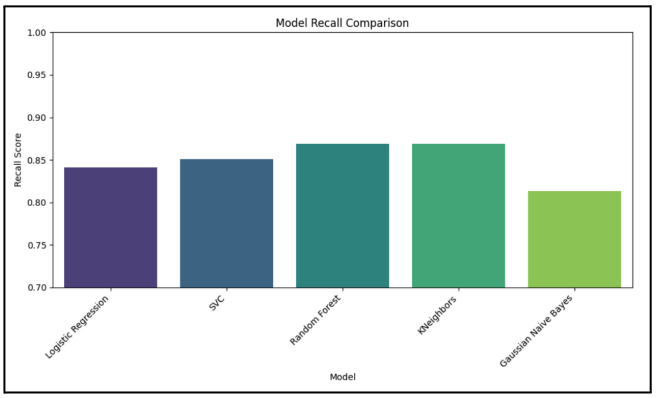


Figure 5. Comparison of recall scores across baseline machine learning models.

2.7 Hyperparameter Optimization

The Random Forest classifier, which demonstrated the highest baseline recall, was selected for optimization. Hyperparameter tuning was performed using **GridSearchCV** with five-fold cross-validation, optimizing for recall. The parameter grid included variations of n_estimators, max_depth, min_samples_split, min_samples_leaf, and class_weight.

The optimal configuration achieved a cross-validation recall of **0.9152**, with parameters including n_estimators = 200, max_depth = 10, and class_weight = 'balanced'.

```
Performing GridSearchCV for Random Forest...
Fitting 5 folds for each of 81 candidates, totalling 405 fits

Best parameters for Random Forest: {'class_weight': 'balanced', 'max_depth': 10, 'min_samples_leaf': 1, 'min_samples_split': 2, 'n_estimators': 200}
Best recall score for Random Forest (CV): 0.9152

Hyperparameter tuning complete for Random Forest.
```

Figure 6. Hyperparameter tuning process using GridSearchCV.

2.8 Model Evaluation and Generalization Analysis

The tuned Random Forest model was evaluated on the held-out test set, achieving an accuracy of 0.8587, precision of 0.8857, recall of 0.8692, F1-score of 0.8774, and ROC-AUC of 0.9212. A confusion matrix was generated to visualize classification outcomes.

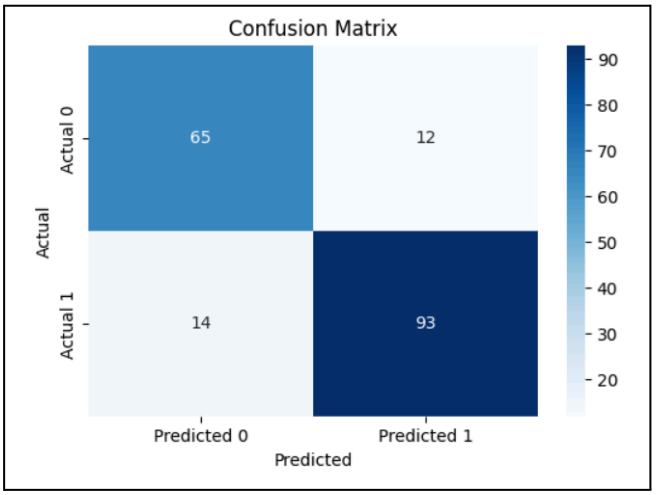


Figure 7. Confusion matrix of the optimized Random Forest model.

To assess generalization, training recall (cross-validation) was compared with test recall. The relatively small gap between these values indicates good generalization and minimal overfitting.

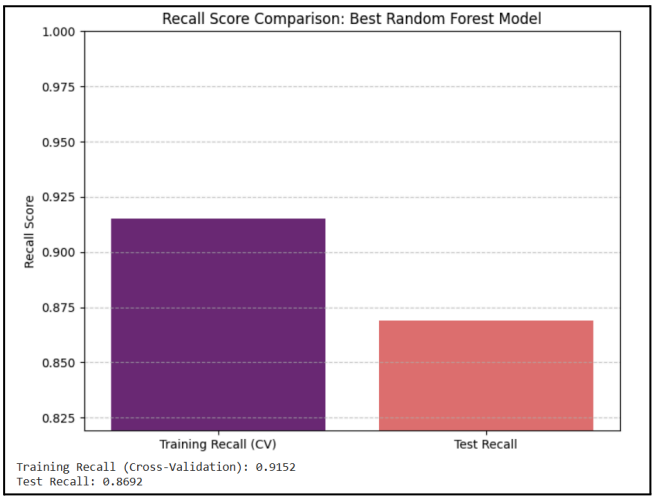


Figure 8. Comparison of training and testing recall values.

2.9 Model Explainability

2.9.1 SHAP Analysis

SHAP (SHapley Additive exPlanations) was applied to analyze global and local feature contributions. Global importance was visualized using SHAP summary plots, while individual predictions were explained using force and waterfall plots.

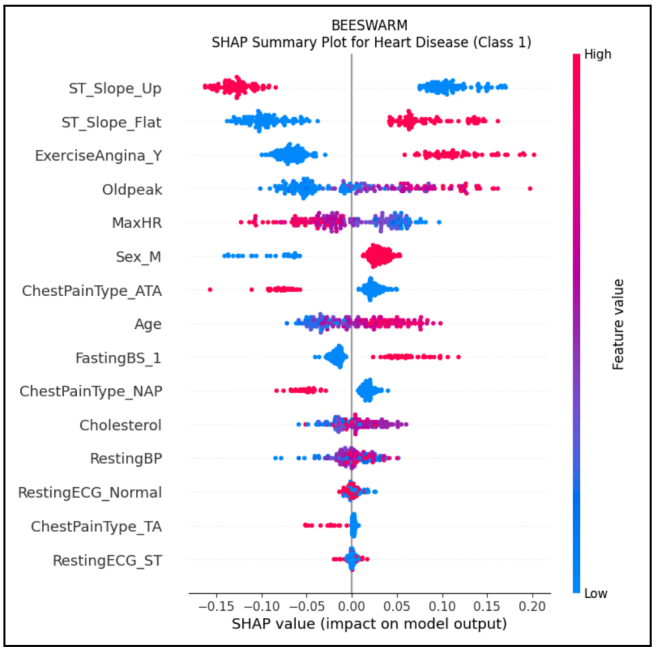


Figure 9. SHAP summary plot showing global feature importance.

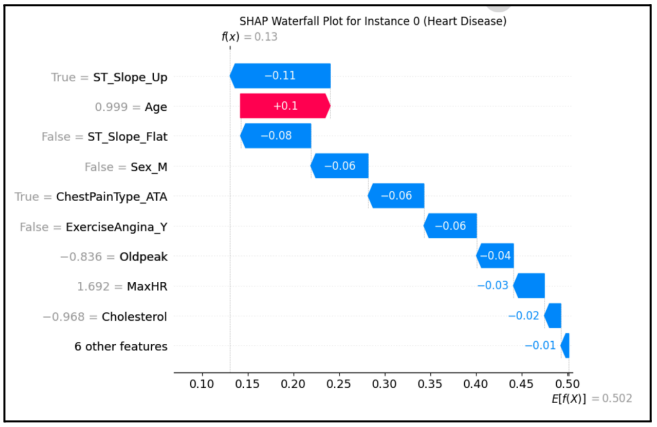


Figure 10. SHAP waterfall plot explaining an individual prediction for heart disease.

2.9.2 LIME Analysis

LIME (Local Interpretable Model-agnostic Explanations) was used to generate instance-level explanations for selected test cases, highlighting the most influential features contributing to individual predictions.

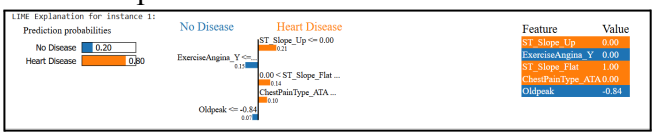


Figure 11. LIME explanation for a representative test instance.

2.10 Out-of-Distribution (OOD) Considerations and Deployment

The machine learning model was trained strictly within the empirical feature ranges observed in the dataset. During deployment, broader **physiologically plausible input ranges** were allowed in the interactive interfaces to

enhance usability. Inputs exceeding the training distribution are considered **out-of-distribution (OOD)** and may yield reduced prediction reliability.

This distinction was explicitly acknowledged in the system design to ensure responsible and transparent deployment, consistent with best practices in medical AI.

2.11 Model Persistence and Interactive Deployment

The final optimized Random Forest model and the fitted StandardScaler were saved using joblib. An interactive prediction interface was implemented using **ipywidgets** for in-notebook validation and **Streamlit** for web-based deployment. The Streamlit application enables real-time predictions with consistent preprocessing and user-friendly interaction.

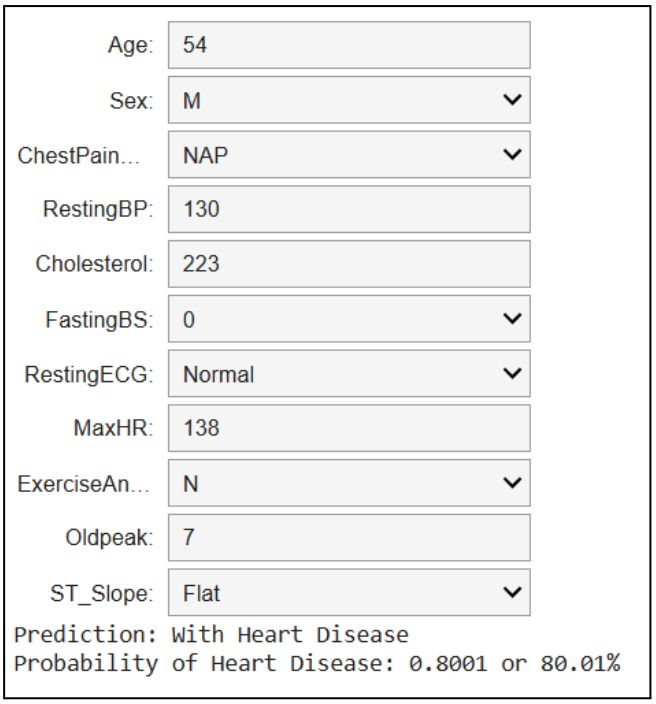


Figure 12. Architecture of the deployed prediction system.

3. Results and Discussion

3.1 Exploratory Data Analysis and Preprocessing Results

Exploratory analysis revealed that although the dataset contained no explicit missing values, medically implausible zero entries were present in Resting Blood Pressure and Serum Cholesterol. Median imputation successfully corrected these anomalies without distorting feature distributions.

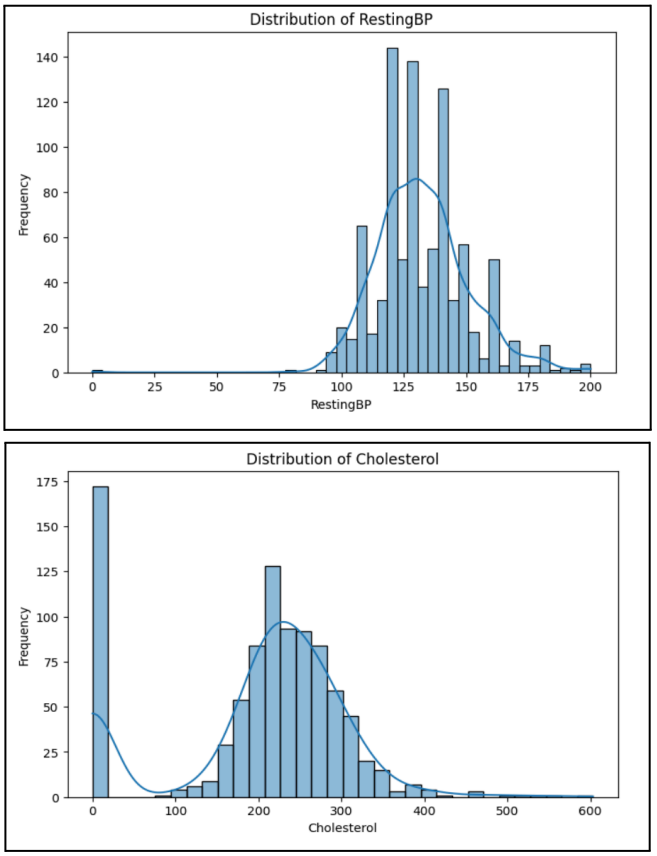


Figure 2. Distribution of numerical features before cleaning.

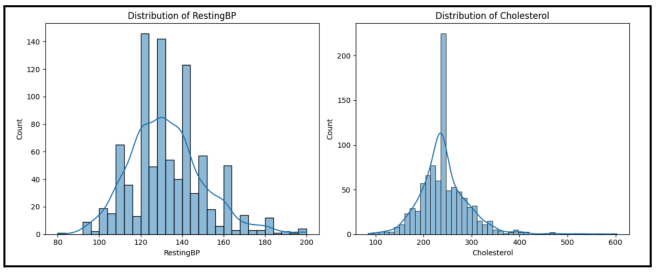
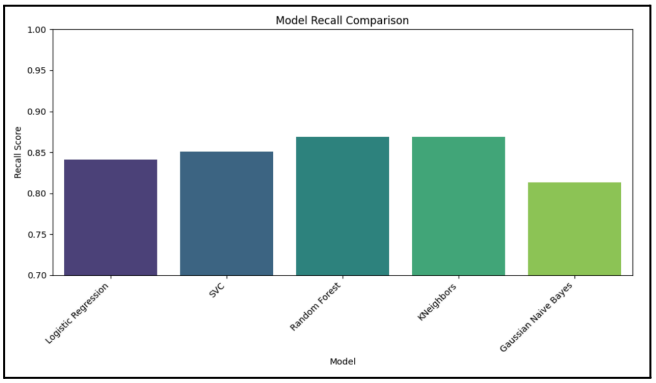


Figure 3. Distribution of RestingBP and Cholesterol after median imputation.

These steps ensured physiologically realistic input data, which is essential for training clinically meaningful predictive models.

3.2 Model Performance Comparison

Among the five evaluated classifiers, the Random Forest model achieved the highest recall on the test set (0.8692), outperforming Logistic Regression, SVM, KNN, and Naïve Bayes. Given the clinical importance of minimizing false negatives, recall was prioritized over accuracy and precision.



Best performing model based on Recall score:

	Model	Recall Score
0	Random Forest	0.8692

Figure 5. Comparison of recall scores across baseline models.

The Random Forest's ensemble structure allows it to model non-linear feature interactions, which likely contributed to its superior performance [1].

3.3 Hyperparameter Optimization and Generalization

```
Performing GridSearchCV for Random Forest...
Fitting 5 folds for each of 81 candidates, totalling 405 fits

Best parameters for Random Forest: {'class_weight': 'balanced', 'max_depth': 10, 'min_samples_leaf': 1, 'min_samples_split': 2, 'n_estimators': 200}
Best recall score for Random Forest (CV): 0.9152

Hyperparameter tuning complete for Random Forest.
```

Figure 6. Hyperparameter tuning process using GridSearchCV.

Figure 6 illustrates the hyperparameter tuning process for the Random Forest classifier using GridSearchCV. A range of values for n_estimators, max_depth, min_samples_split, min_samples_leaf, and class_weight were systematically evaluated with 5-fold cross-validation, optimizing for recall to ensure the model generalizes effectively to unseen data.

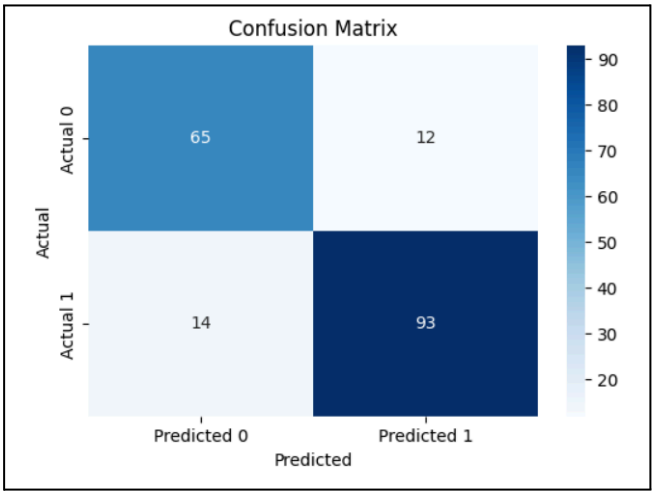


Figure 7. Confusion matrix of the tuned Random Forest model.

Figure 7 presents the confusion matrix of the optimized Random Forest model, summarizing its classification performance on the test dataset by comparing predicted outcomes against true clinical labels. This matrix provides a detailed breakdown of correct and incorrect predictions, offering insight into the model's diagnostic behavior and error characteristics.

The model correctly identified **65 individuals as not having heart disease (True Negatives)** and **93 individuals as having heart disease (True Positives)**, demonstrating strong overall discriminatory capability. However, **12 cases were incorrectly classified as heart disease when no disease was present (False Positives)**, representing Type I errors, while **14 cases were incorrectly classified as disease-free when heart disease was actually present (False Negatives)**, representing Type II errors.

From a clinical perspective, false negatives are particularly critical, as they correspond to missed diagnoses that may delay treatment and increase patient risk. Despite this, the relatively low number of false negatives indicates that the model maintains a favorable balance between sensitivity and specificity. Overall, the confusion matrix confirms that the optimized model performs effectively as a screening tool, with a strong emphasis on correctly identifying patients with heart disease while keeping diagnostic errors at an acceptable level.

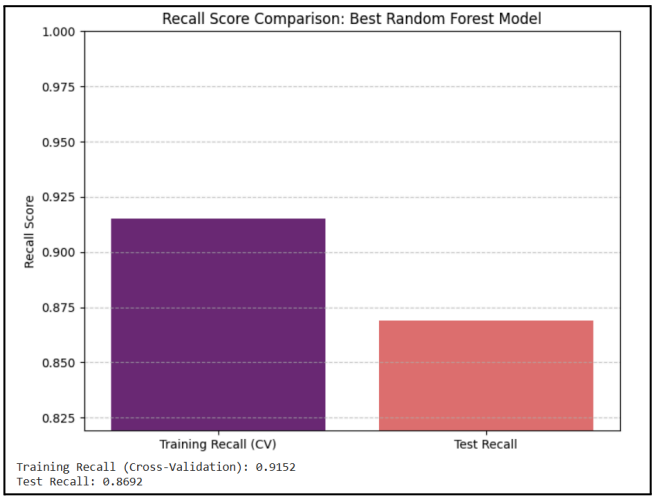


Figure 8. Comparison of training and testing recall values.

Hyperparameter tuning via GridSearchCV improved cross-validation recall to 0.9152. Evaluation on the held-out test set yielded a recall of 0.8692, indicating good generalization with

minimal overfitting.

3.4 Explainability Analysis

To address the interpretability limitations commonly associated with ensemble machine learning models, this study employed both SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations). The combined use of these techniques enables a dual-level explanation of model behavior: **global interpretability** through SHAP and **local, instance-specific reasoning** through LIME. This complementary approach enhances transparency, trust, and clinical relevance.

3.4.1 Global Interpretation Using SHAP

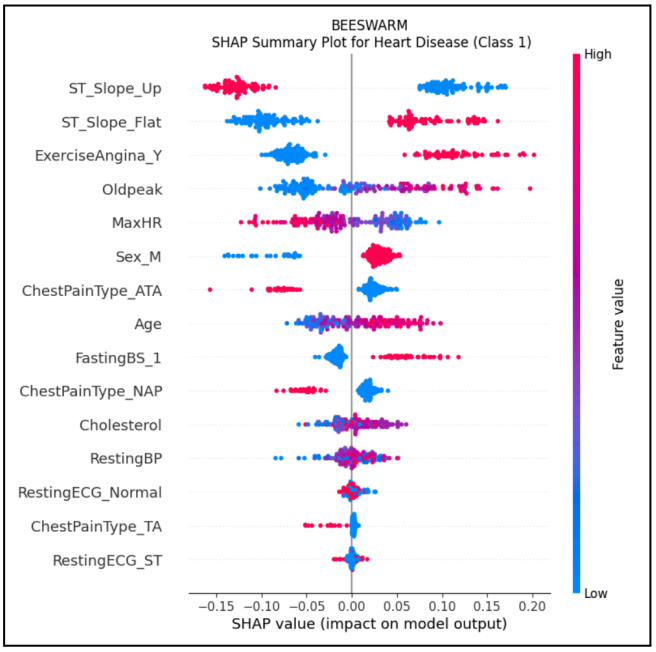


Figure 9. SHAP summary plot illustrating global feature importance.

SHAP (SHapley Additive exPlanations) was employed to analyze the global behavior of the optimized Random Forest model. Figure 9 presents the SHAP beeswarm summary plot for **Heart Disease (Class 1)**, illustrating both the relative importance of features and the direction and variability of their impact across the test dataset.

In the beeswarm plot, features are ranked vertically according to their mean absolute SHAP values, with higher-ranked features exerting greater overall influence on the model's predictions. The most influential variables include **ST_Slope_Up**, **ST_Slope_Flat**, **ExerciseAngina_Y**, **Oldpeak**, and **MaxHR**, highlighting the dominant role of electrocardiographic and exercise-related indicators in heart disease prediction.

A key strength of the SHAP beeswarm visualization is its ability to convey the **direction of feature impact** through color encoding. Each point represents an individual observation, where **red points indicate high feature values** and **blue points indicate low feature values**. The horizontal position of each point reflects the SHAP value, showing whether that feature pushes the prediction toward or away from heart disease:

- **Red points to the right** indicate that a **high feature value increases** the predicted likelihood of heart disease.
- **Blue points to the right** indicate that a **low feature value increases** the predicted likelihood of heart disease.
- **Red points to the left** indicate that a **high feature value decreases** the predicted likelihood of heart disease.
- **Blue points to the left** indicate that a **low feature value decreases** the predicted likelihood of heart disease.

Additionally, the **horizontal spread of points** for each feature reflects the **distribution of SHAP values** across patients. A wider spread indicates greater variability in the feature's influence, suggesting interaction effects and non-linear relationships captured by the ensemble model.

Several clinically interpretable patterns are evident from the plot. **ST_Slope_Up** shows red points concentrated on the negative SHAP axis, indicating that the presence of an up-sloping ST segment consistently reduces the predicted risk of heart disease. In contrast, **ST_Slope_Flat** exhibits red points predominantly on the positive SHAP axis, confirming that a flat ST segment strongly increases disease likelihood. Similarly, **ExerciseAngina_Y = 1** (red points) pushes predictions toward heart disease, while its absence is associated with lower risk.

For **Oldpeak**, higher values (red points) are largely positioned on the positive SHAP axis, indicating increased risk associated with greater ST depression during exercise. **MaxHR** displays an inverse pattern, where lower values (blue points) contribute positively to heart disease prediction, while higher values are associated with negative SHAP values, reflecting better cardiovascular performance. **Age** shows a gradual

shift, with higher values generally increasing predicted risk, though with moderate dispersion, suggesting interaction with other clinical variables.

Lower-ranked features such as **Cholesterol**, **RestingBP**, and **RestingECG-related variables** exhibit smaller SHAP magnitudes clustered near zero, indicating limited but non-negligible contributions in specific cases.

Overall, the SHAP analysis demonstrates that the model's predictions are driven primarily by clinically meaningful electrocardiographic and exercise-response features. The consistent alignment between SHAP-derived patterns and established cardiovascular risk factors supports the validity, transparency, and medical plausibility of the model's global decision-making behavior.

3.4.2 Local Interpretation Using SHAP Waterfall Plot

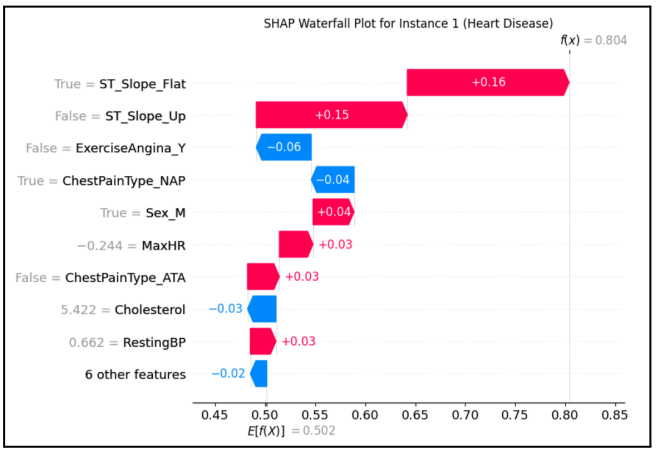


Figure 10. SHAP waterfall plot explaining an individual prediction for heart disease.

Figure 10 presents the SHAP waterfall plot for **Instance 1**, illustrating how individual feature contributions incrementally shift the model's prediction from the baseline probability of heart disease ($E[f(X)] \approx 0.502$) to the final predicted probability ($f(x) \approx 0.804$). This substantial upward shift indicates a **high-risk classification** for heart disease.

In the waterfall plot, **positive SHAP values (red bars)** represent features that increase the predicted likelihood of heart disease, while **negative SHAP values (blue bars)** indicate protective features that reduce risk. The magnitude of each bar reflects the strength of the feature's influence on the prediction.

The strongest contributors driving the prediction **toward heart disease** are electrocardiographic features related to ST-segment morphology. The presence of a **flat ST segment (ST_Slope_Flat = True)** is the most influential factor, producing the largest positive shift in the predicted probability. Clinically, a flat ST segment is a well-established marker of myocardial ischemia and abnormal cardiac stress response. This effect is further reinforced by the **absence of an up-sloping ST segment (ST_Slope_Up = False)**, which removes a key protective ECG indicator and significantly increases predicted risk.

Additional risk-increasing contributions arise from **male sex**, **elevated resting blood pressure**, **lower maximum heart rate**, and the **absence of atypical angina (ChestPainType_ATA = False)**. These factors are consistent with known cardiovascular risk profiles, where reduced exercise tolerance, hemodynamic strain, and male sex are associated with increased disease prevalence.

Several features act in the opposite direction, partially mitigating risk. The **absence of exercise-induced angina (ExerciseAngina_Y = False)** and the presence of **non-anginal chest pain (ChestPainType_NAP = True)** contribute negatively to the prediction, reflecting symptom patterns that are generally less indicative of ischemic heart disease. Additionally, **lower cholesterol levels** and the combined effect of other minor features provide modest protective influence.

Despite these mitigating factors, their cumulative effect is insufficient to offset the dominant electrocardiographic and physiological risk indicators. As a result, the additive combination of feature contributions yields a final predicted probability substantially above the population baseline, leading the model to classify this instance as **high risk for heart disease**.

Overall, this waterfall plot demonstrates that the model's prediction for Instance 1 is driven primarily by **objective ECG abnormalities**, with secondary contributions from demographic and hemodynamic factors. The explanation highlights how strong ischemic indicators can outweigh protective symptoms, reinforcing the clinical plausibility of the model and illustrating its ability to balance competing

signals in an interpretable and medically coherent manner.

3.4.3 Local Interpretation Using LIME

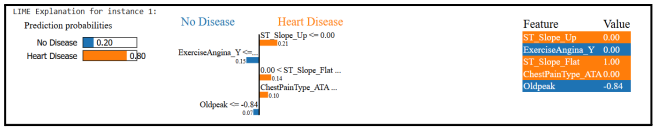


Figure 11. LIME explanation for a representative test instance.

While SHAP provides a dataset-level explanation, LIME was used to interpret individual predictions at the patient level. Figure 11 presents the LIME explanation for a representative test instance (Instance 1), for which the model predicted **Heart Disease with a probability of 0.80**.

The LIME visualization decomposes the prediction into feature-level contributions, indicating which variables pushed the model toward or away from the predicted class. For this instance, several electrocardiographic features strongly influenced the positive heart disease classification.

The most influential contributors toward **Heart Disease** were **ST_Slope_Up ≤ 0.00** and **ST_Slope_Flat = 1.00**, indicating the absence of an up-sloping ST segment and the presence of a flat ST slope. Clinically, a flat ST segment is a well-established marker of myocardial ischemia and abnormal cardiac stress response, which strongly elevates cardiovascular risk. Additionally, **ChestPainType_ATA = 0.00** contributed positively to the prediction, suggesting that the chest pain profile corresponds to types more strongly associated with heart disease.

Conversely, **ExerciseAngina_Y = 0.00** and **low Oldpeak (≤ −0.84)** acted as protective factors, slightly reducing the predicted risk. The absence of exercise-induced angina and minimal ST depression during exercise are generally associated with lower ischemic burden. However, these protective effects were outweighed by the stronger influence of electrocardiographic abnormalities.

Overall, the LIME explanation demonstrates that the model prioritizes **ECG-related indicators over symptom-based features** when making patient-level predictions. This behavior is clinically consistent, as electrocardiographic findings often provide more objective evidence of underlying cardiac pathology than self-reported symptoms alone.

3.5 Out-of-Distribution (OOD) Inputs and Deployment Considerations

The model was trained strictly within the empirical feature ranges present in the dataset. However, the deployed interfaces accept broader physiologically plausible ranges to improve usability. Inputs falling outside the training distribution are considered out-of-distribution and may yield reduced prediction reliability.

This design choice balances accessibility and methodological integrity, aligning with best practices in responsible AI deployment [1].

4. Limitations and Future Work

Despite promising results, this study has several limitations. The dataset represents a limited population and lacks lifestyle variables such as smoking status, body mass index, and family history, which are known cardiovascular risk factors. Additionally, the model does not incorporate longitudinal or imaging data.

Future work should include validation on larger, multi-center datasets, incorporation of additional clinical and lifestyle features, and calibration techniques to quantify prediction uncertainty, particularly for out-of-distribution inputs.

5.Conclusion and Recommendations

This study demonstrates that a tuned Random Forest model, combined with explainable AI techniques, can effectively predict heart disease using structured clinical data while maintaining transparency and generalization. The integration of SHAP and LIME provides clinically meaningful insights into model behavior, enhancing trust and interpretability.

It is recommended that such models be used as supportive screening tools rather than standalone diagnostic systems. With further validation and dataset expansion, the proposed framework has strong potential for integration into clinical decision support systems.

6. References

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